MILITARY SPECIFICATION

BATTERY CELLS, MAIN STORAGE, LEAD-ACID, DEEP SUBMERGENCE VEHICLES (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers lead-acid cells comprising the main storage batteries for deep submergence vehicles (DSV).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-S-801 - Sulfuric Acid, Electrolyte; for Storage Batteries.
PPP-B-587 - Boxes, Wood, Wirebound, Pallet Type.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-650 - Crates, Wood, Open and Covered.

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MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

PUBLICATIONS

MILITARY

NAVSEA 0901-LP-223-0000 - Naval Ships Technical Manual, Chapter 223, Lead-Acid Submarine Storage Batteries.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements.

3.1.1 Design and construction. Cells (see 6.3.1) shall be designed to function specifically in an oil-sulfuric acid environment under pressure-compensated conditions, but the performance of the cells shall not be impaired when operated at standard atmospheric conditions and without oil. Each cell shall be 270 ampere-hours (Ah) capacity at the 6-hour discharge rate at 25°C ± 3°C. Installation shall consist of 90 cells arranged in four batteries in two boxes: one group of 30 series-connected cells (60-volt system) and one group of 15 series-connected (30-volt system) per battery box. The two battery boxes shall be operated in parallel.

3.1.2 Installation and interface. Each battery box shall be installed external to the ship pressure hull. The cells shall be open to the pressure-compensating oil (see 3.3.10) filling the battery box and serving as a buffer between the electrolyte and the seawater.

3.2 Electrical requirements.

3.2.1 Life. The cell, when installed, shall be capable of delivering 300 equivalent cycles (an equivalent cycle is 300 Ah of charge) or 36 months whichever occurs first. The guaranteed life of the cells shall be considered at an end when the cells cannot be made to deliver 80 percent of their rated capacity (see 6.2).

3.2.2 Capacity. The rating of each cell, based on an initial electrolyte temperature of 25°C ± 3°C and specific gravity of 1.290 ± 0.010 shall be not less than that shown in table I.

TABLE I. Electrical rating.

Time (hours) (h)	Discharge current amperes (A)	Capacity (Ah)	Minimum cell final voltage (volts) (V)	Battery (series) minimum final voltage (V)	
				15-cell	30-cell
0.5	280	140	1.26	21.9	43.8
1	170	170	1.43	23.5	47.0
3	70	210	1.66	25.1	50.2
5	50	250	1.72	25.6	51.3
6	45	270	1.74	25.9	51.9
10	28	280	1.76	26.3	52.6
20	16	320	1.79	26.8	53.6

3.2.3 Charge.

3.2.3.1 Full charge. Cells shall be fully charged when three consecutive hourly readings of voltage or specific gravity of the electrolyte, or both, corrected to 25°C, show no increase.

3.2.3.2 Charging. Cells or battery shall be charged at a charging rate or rates until three consecutive voltage readings remain constant at hourly readings. However, the total input in Ah shall not exceed 135 percent of the previous discharge. The temperature of the electrolyte during charging period shall not exceed 50°C. Charge shall be defined as a function of the electrolyte temperature of the pilot cell for each of the four batteries (see 3.1.1).

3.2.4 Electrical breakdown. Cells, completely assembled and in a dry condition, shall not conduct more than 50 microamperes of current between the positive and negative terminals when tested as specified in 4.4.1.2.

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3.3 Physical requirements.

3.3.1 Cell dimensions. Cell dimensions shall be as shown on figure 1. The overall dimensions of the cell in millimeters (mm) shall be:

	Maximum
Width	160
Length	110
Height[1/]	460

3.3.2 Cell mass. Any dry cell, complete with entrainment eliminator, nuts and bolts, shall not exceed 15.6 kilograms (kg) while the average shall not exceed 15.1 kg. The mass of the complete cell with electrolyte shall not exceed 20.0 kg.

3.3.3 Cell jar. Cell jars shall be resistant to sulfuric acid, seawater, and dielectric oil with dimensions as shown on figure 1. Jars shall meet the dielectric requirements defined in NAVSEA 0901-LP-223-0000, chapter 223.

3.3.4 Cell cover. Cell covers (and method of sealing covers to cell jars) shall be compatible with the sulfuric acid/sea water/dielectric oil environment specified in 3.3.3 under the additional environmental requirements specified in 3.4. Cell covers shall be designed and furnished with an electrolyte entrainment eliminator which shall be the only cover fitting allowing the passage of gases from the cell.

3.3.5 Electrodes.

3.3.5.1 Active materials. Lead alloys and active materials used in the manufacture of the cell electrodes shall not contain impurities exceeding the limitations shown in table II. Reclaimed materials shall be used to the maximum extent possible.

TABLE II. Material impurities.

Material	Iron	Copper	Arsenic	Platinum	Bismuth
Lead alloy, maximum percent by mass	0.012	0.050	0.500	0.00001	0.050
Active material, maximum percent by mass	0.020	0.005	0.00001	0.0001	0.100

3.3.5.2 Physical characteristics. The physical characteristics of the electrode pack assembly shall be as described for the quality conformance test samples and shall be submitted at the time of quality conformance testing. They shall include the following (for the Government only):

(a) Density of active material (after electrochemical formation and drying of the paste).
(b) Electrode dimensions, including thickness.
(c) Mass of active material per electrode and mass of electrodes.
(d) Number of positive and negative electrodes.
(e) Separator system description from positive electrode to negative electrode (see 3.3.6).
(f) Separator characteristics in terms of thickness, porosity, and average pore size (see 3.3.6).

3.3.6 Separators. Separators used in the cell shall have the following characteristics:

Porosity: 50 percent minimum
Electrolyte resistivity in sulfuric acid of 1.280 specific gravity: 6.4 ohm-cm maximum
Moisture content by mass: not to exceed 2 percent in a maximum of 75 percent relative humidity at a maximum temperature of 35°C

1/ Includes terminal posts and electrolyte entrainment eliminator device.

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Impurities: not to exceed the following limits by mass:

Aluminum	0.1 percent
Copper	0.06 percent
Iron	0.06 percent
Magnesium	0.06 percent
Manganese	0.06 percent
Chloride (as Cl):	0.006 percent

Total metallics other than aluminum, iron, and manganese not to exceed 0.073 percent by mass

3.3.7 Cell terminals. Terminal bolts, nuts, and washers used for terminal connections shall be made of a stainless steel which is resistant to the specified electrolyte. Cable connectors shall be lead-coated copper. Lead coating on parts not exposed to direct action of electrolyte shall be not less than 0.1-mm thick. The design of any individual terminal shall be such that the temperature rise (above the starting electrolyte temperature) at the top of the terminal does not exceed 15°C at anytime during a discharge at the 1-hour rate. Only one terminal for each polarity shall be used. Terminal posts shall be marked as specified in 3.3.12.1.

3.3.8 Intercell connectors and jumpers. Intercell connectors shall be lead.

3.3.9 Electrolyte.

3.3.9.1 Composition. Sulfuric acid used for preparing the electrolyte shall meet the impurity requirements of O-S-801. Electrolyte specific gravity when fully charged, shall be 1.290 ± 0.010. Unless otherwise specified (see 6.1.1), the cells shall be furnished without electrolyte.

3.3.9.2 Level. The manufacturer shall provide a means by which the electrolyte level may be determined, without removing the cell from the battery box, for the purpose of electrolyte adjustment.

3.3.9.3 Entrainment eliminator. Each cell shall be equipped with an electrolyte entrainment eliminator device. The device shall separate the gas from the electrolyte bubbles permitting the electrolyte droplets to return into the cell and the gas to exit.

3.3.9.4 Electrolyte impurities. The amount of impurities in the electrolyte shall not exceed the limits specified in table III at any time before or during cycle life.

TABLE III. Electrolyte impurities.

Impurity	Determined as	Maximum percent by mass
Ammonia	NH_4	0.001
Antimony	Sb	0.001
Arsenic	As	0.0001
Chlorine (including perchlorates)	Cl	0.012
Copper	Cu	0.005
Iron	Fe	0.012
Manganese	Mn	0.00006
Nickel	Ni	0.0001
Nitrates	NO_3	0.002
Platinum	Pt	0.00001
Suspended matter	Lead compounds only	--------
Inert matter	(e.g., glass)	0.010

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3.3.10 Compensating oil. A compensating oil shall be used in the battery box and cells. The oil shall be utilized to fill the upper part of each cell and all voids in the battery box in order to distribute the pressure of the sea equally on all components within the battery. The oil shall be entirely unaffected by and nonmiscible in seawater and electrolyte and shall maintain a low viscosity at deep ocean and storage temperatures. A naphthene based "white" mineral oil, equivalent to EXXON Corporation, MARCOL 70, is preferred for use. A small amount of "Oil Blue N" dye (methylamino-butylamino anthraquinone) is sometimes added to make it readily distinguishable from electrolyte or other types of oil.

3.3.11 Pressure-tightness. Cells shall withstand 7 kilopascals (kPa) air pressure for 1 minute without loss of pressure.

3.3.12 Cell identification.

3.3.12.1 Serial numbers. Manufacturer's serial numbers shall be visible from the top of the installed cells. The serial number shall consist of five digits, the first two representing the year of manufacture and the last three the manufacturer's serially assigned number. Markings shall be applied in a manner not easily effaced. Positive terminal posts shall be marked "Pos" or "+". Negative terminal posts shall be marked "Neg" or "-".

3.3.13 Mercury. Mercury shall not be used in the manufacture or in any test in which there is any danger of introducing mercury into the cell through accident or otherwise.

3.3.14 Protecting baffles. A horizontal perforated protecting baffle, at least 1 mm-thick, shall be securely held in place in each cell above the plates and separators. The baffle shall be so placed as to prevent injury to the separators due to instruments inserted through the filling apertures in the cell cones. The baffle shall not rest or bear on the plates.

3.4 Environmental requirements.

3.4.1 Temperature. Cell shall operate, with a pressure compensating oil (see 3.3.10) occupying the space above the electrolyte in the cell, for extended periods, at temperatures from minus 1°C to 50°C. Cell shall be suitable for exposure to air temperatures from minus 30°C to 55°C.

3.4.2 Pressure. Cell shall operate, with a pressure compensating oil (see 3.3.10) occupying the space above the electrolyte in the cell for extended periods in a compensated pressure environment as high as 20 megapascals (MPa).

3.4.3 Angle of operation. Cell shall withstand 45 degree angles of operation continuously without physical or electrical damage. Electrolyte shall not spill nor shall oil come in contact with electrodes when the cells are tilted 45 degrees in any direction.

3.5 Accessories. Accessories (see 6.3.4) shall be delivered with each set of 90 cells in sufficient quantity to meet the manufacturer's design for the physical and electrical installation of the cells in the two battery boxes (see 3.1.1).

3.6 Engineering services (see 6.1.1). Services of the contractor's engineer shall be available upon the Naval Ship Engineering Center (NAVSEC) request in order to:

(a) Assist the shipyard in preparing the cells for installation and installing them in the battery boxes.
(b) Assist the ship's force in becoming familiar with the cells.
(c) Take corrective action in case of unexpected problems.
(d) The contractor shall visit the ship and inspect the cell condition at least once per quarter.
(e) Participate in technical meetings with NAVSEC representatives or the ship's force with regard to the cells and in locations other than the manufacturer's plant.

A maximum of 20 mandays will be allowed for the entire life of each set of cells (see 6.3.2) purchased on the same contract (see 6.4). A field engineering report shall be prepared in accordance with the data ordering document included in the contract or order (see 6.1.2).

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3.7 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order and as specified in 3.7.1 and 3.7.2.

3.7.1 Drawings. The contractor shall prepare engineering drawings in accordance with the data ordering document included in the contract or order (see 6.1.2) for review by NAVSEC. As a minimum, the following drawings shall be included, with no mention of proprietary information or any other document not supplied:

(a) Cell case.
(b) Cell cover.
(c) Positive electrode.
(d) Negative electrode.
(e) Separators.
(f) Terminal.
(g) Terminal-cover assembly.
(h) Cell control drawing.
(i) Cell pack assembly.
(j) Entrainment eliminator.
(k) Sketch showing relative positions of tops of plates and separators with respect to cover underside as well as level of electrolyte.
(l) Data list (list of all drawings that are used in the design and manufacture of the cells).

3.7.2 Service manual. The contractor shall prepare a service manual in accordance with the data ordering document included in the contract or order (see 6.1.2) and the following:

(a) The manual describing battery operation (see 3.1.1) shall not be in conflict with the contents of NAVSEA 0901-LP-223-0000, chapter 223. The service manual shall make reference to specific articles of the technical manual, wherever applicable.
(b) The manual shall contain a parts list for onboard repair.
(c) The manual shall contain charge/discharge data including temperature/voltage/gassing (TVG) curves for charging, and discharge curves at standard discharge rates.
(d) When an applicable manual has been previously submitted, a certificate of prior submittal will be acceptable unless otherwise specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system as specified and in accordance with the data ordering document included in the contract or order (see 6.1.2).

4.1.2 Test reports. Test reports covering data from quality conformance testing shall be submitted to the Defense Contract Administration Service (DCAS) enabling the authorization cited in 4.2.2. The report shall be prepared in accordance with the data ordering document included in the contract or order (see 6.1.2).

4.2 Quality conformance inspection.

4.2.1 Sampling. Six sample cells from the production lot shall be randomly selected for the quality conformance inspection specified in 4.2.2.

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4.2.2 Inspection. Sample cells, selected in accordance with 4.2.1, shall be subjected to the examination specified in 4.3 and the tests specified in 4.4.1 through 4.4.2.7. Any sample cells, as well as the set represented by the sample, which fail to pass the required tests, shall be rejected. Upon successful completion of these required tests, the DCAS will authorize shipment of the six sample cells to a laboratory as specified by NAVSEC (see 6.1.1), for further testing specified in 4.4.3 and 4.4.4. At the same time, the set represented by the sample shall be shipped to destination.

4.3 Examination. Parts used in the assembly of the cells shall be examined for conformance to the requirements of this specification and to the manufacturer's quality control provisions. Assembled cells shall be examined for any visible physical defects and for conformance to the requirements of this specification which do not involve tests. Any defective cell shall be rejected.

4.4 Tests.

4.4.1 Tests for all cells manufactured.

4.4.1.1 Pressure-test. All assembled cells shall be air-pressure tested at 7 kPa for 1 minute through an opening of the cover with all other openings sealed and shall show no leak or drop in pressure.

4.4.1.2 Dry cell test. The cell in the dry condition shall be tested for shorts by applying an electrical potential of 1000 volts between the positive and negative terminals. The current measured shall not exceed 50 microamperes.

4.4.2 Tests on sample wet cells.

4.4.2.1 Mass. The mass of the sample cells shall meet the requirements specified in 3.3.2.

4.4.2.2 Capacity and voltage determination. After a maximum of three conditioning cycles in accordance with the manufacturer's recommendation, each sample cell shall be discharged as specified in 4.4.2.7. The discharges shall be made to the voltage cut-off specified in 3.2.2. In all cycles, each cell individually shall meet the requirements of 3.2.2 and 3.2.3. If a cell fails any cycle, the cycle may be repeated not over two times for that particular cell.

4.4.2.3 Charge acceptance. Each sample cell shall be tested for charge acceptance to determine conformance with 3.2.3.2 following any discharge rate. The requirement of 3.2.3.1 may be demonstrated in conjunction with this test.

4.4.2.4 Angle of operation. Each sample cell, within 1 hour after ending any charge, shall be tilted in any direction 45 degrees from vertical without spilling electrolyte. Cells shall be tilted 60 degrees from vertical or to an angle less than 60 degrees where spill occurs to determine the maximum inclination capability of the manufacturer's design.

4.4.2.5 Electrolyte level indication. The manufacturer's electrolyte level indicator shall be tested by determining the electrolyte level for various levels which occur during charged, discharged, and open circuit status or which may occur during abnormal conditions (too high or too low).

4.4.2.6 Terminal temperature. Sample cells shall be connected as a battery and insulated on four sides so as to approximate an adiabatic condition as closely as possible. The temperatures of the positive terminal and electrolyte shall be measured by thermocouples or thermistors and recorded at 15-minute intervals during a 1-hour rate discharge with an electrolyte starting temperature of 25°C ± 1°C and an ambient temperature as specified in 4.4.3. Cells shall meet the requirement specified in 3.3.7.

4.4.2.7 Cycling under pressure. Sample cells shall meet the electrical requirements applicable to the following tests:

(a) Cycle 1. After filling, the cells shall be charged in accordance with the procedure recommended in the service manual, then placed in an oil-filled vessel, pressurized to 20 MPa at ambient temperature and discharged at 175 A to the cut-off voltage.

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(b) Cycle 2. Cells shall be depressurized at the rate of 200 kPa to atmospheric pressure, then immediately recharged at the existing electrolyte and ambient temperatures, overcharged at the same finishing rate for 3 hours, then immediately pressurized to 20 MPa. Cells shall then be discharged at 45 A to the cut-off voltage.

(c) Cycle 3. Cells shall be depressurized at the rate of 200 kPa to atmospheric pressure.

4.4.3 Test discharges at various rates. Sample cells shall be subjected to a series of discharges in the following order: 10-hour rate, 5-hour rate, 20-hour rate, 1-hour rate. Rates and limiting voltages shall be as specified in 3.2.2. The initial electrolyte temperature shall be 25°C ± 5°C (room temperature). The electrolyte temperature shall not exceed 55°C at any time during the discharge. Ninety percent of the service rating at each discharge rate shall be demonstrated before the voltage cut-off or the limiting temperature is reached within the first ten cycles.

4.4.4 Life test. Sample cells shall be tested for life in accordance with the instructions specified by NAVSEC to verify conformance with the electrical, physical, and environmental requirements of 3.2, 3.3, and 3.4. The life cycling test shall be a series of cycles at the discharge rates specified in 4.4.3 and repeated consecutively until 300 equivalent cycles are exceeded, except the 1/2-hour rate which shall be performed every 50 actual cycles.

4.5 Inspection of preparation for delivery. The preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging. Preservation-packaging shall be level A as specified herein.

5.1.1 Level A.

5.1.1.1 Cells. Cells shall be individually unit protected by method III in accordance with MIL-P-116 and packed as specified in 5.2.2.1. When wet cells are specified (see 3.3.9.1), each cell prior to packaging shall be placed in a minimum 0.1 mm plastic bag. Bag closure shall be at the option of the contractor except that bare wire ties or staples shall not be used.

5.1.1.2 Spare cells, parts, and accessories. Spare and test cells shall be packaged as specified in 5.1.1.1. Spare parts shall be individually packaged in accordance with the guidelines of MIL-P-116 except that items used in quantities greater than one or in sets may be packaged together.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1.1).

5.2.1 Level A. Cells, spare parts, and accessories, packaged as specified in 5.1.1, shall be packed as specified in 5.2.2, except that shipping containers shall be of the level A or overseas type or class of the container selected.

5.2.2 Level B.

5.2.2.1 Cells. Cells, packaged as specified in 5.1.1, shall be packed in containers as follows:

Installation cells - maximum of 30 cells per container
Spare cells - 9 cells per container
Test cells - 6 cells per container

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5.2.2.1.1 Containers. In the design of the container, the following specifications provide guidance for construction, assembly, and materials:

(a) PPP-B-587
(b) PPP-B-601
(c) PPP-B-621
(d) PPP-C-650
(e) MIL-C-104

The following criteria shall be incorporated in the container design:

(a) Containers shall accept either filled or empty cells.
(b) Cells shall be segregated within the container by separators, blocking, or bracing.
(c) Container sides and ends shall have structural members to prevent container distortion.
(d) Container tops, except wirebound types, shall be secured by screw type fasteners in lieu of nailing.
(e) Containers shall be designed for reusability.

5.2.2.2 Spare parts and accessories. Spare parts and accessories, packaged as specified in 5.1.1.2, shall be packed in fiberboard boxes conforming to PPP-B-636, domestic grade or PPP-B-640, class 1. Container type, style, and variety shall be at the option of the contractor. Container closure and reinforcing as applicable, shall be in accordance with the appendix to the applicable container specification with method I closure applicable to PPP-B-636 boxes.

5.2.2.3 Consolidated packing. Items specified in 5.2.2.2 when supplied with a complete set of cells, may be consolidated within a single shipping container. The container(s) shall be provided with two index lists completely itemizing and identifying the container contents. One copy shall be placed inside the container in a location offering ease of access and removal. One copy shall be attached to the exterior side or end of the container in a protected location which will prevent loss or damage during shipment and storage. Lists shall be enclosed in a waterproof sealed bag or envelope.

5.2.3 Level C. Cells, spare parts, and accessories shall be packed as specified in 5.2.2.

5.3 Marking. In addition to any special marking required (see 6.1.1), or herein, interior packages (unit and intermediate) and shipping containers shall be marked in accordance with MIL-P-116 and MIL-STD-129 as applicable to the levels of packaging and packing specified.

5.3.1 Special markings.

5.3.1.1 Cell containers. Cell containers shall be marked with the following:

(a) Fragile and up-arrows (see MIL-STD-129).
(b) Manufacturer's storage instructions.
(c) Manuals enclosed (see 5.4.2).
(d) Labelling as required by Department of Transportation (for cells shipped wet and charged).
(e) Structural and handling markings such as center-of-balance and lift points.

5.4 Technical data (see 3.7).

5.4.1 Drawings. Drawings shall be prepared for shipment in a manner to afford protection against deterioration, damage or loss during shipment from the supply source to the receiving activity. Containers, packing, or method of shipment shall as a minimum comply with postal regulations or carrier rules as applicable to the mode of transportation.

5.4.2 Technical manuals. Technical manuals shall not be directly shipped by the manufacturer with a battery. After NAVSEC concurrence, bulk manuals shall be prepared for shipment in accordance with the applicable manual specification (see 6.1.1) and shipped to NAVSEC for distribution.

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6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When cells shall be furnished with electrolyte (see 3.3.9.1).
(c) If engineering services are required (see 3.6).
(d) Address for shipment of sample cells to a test laboratory (see 4.2.2).
(e) Level of packing required (see 5.2).
(f) Special marking (see 5.3).
(g) Preparation for delivery requirements for bulk manuals (see 5.4.2).

6.1.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.6	Field engineering reports	UDI-E-23127	----------
3.7.1	Drawings, engineering and associated lists, level 3 (production)	DI-E-7015	Design activity designation - contractor Drawing number - contractor Parts list and data list - MIL-STD-100 Delivery of hard copy - NAVSEC
3.7.2	Manual, technical, standard, basic issue	DI-M-2044	Type I of MIL-M-15071
4.1.1	Inspection system program plan	DI-R-4803	----------
4.1.2	Test reports	DI-T-2072	----------

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.1.2.1 The data requirements of 6.1.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.2 Guarantee. The guarantee shall be as specified in 6.2.1 through 6.2.6.

6.2.1 The contractor guarantees the life of the cells, furnished under contract for 300 equivalent cycles (an equivalent cycle is 300 Ah of charge) or 36 months whichever occurs first.

6.2.2 The period of the guarantee for cells furnished dry and uncharged shall be considered to begin on the date the cells are filled, except when the time between the date of shipment and the date of initial charging exceeds 2 years, in which case the guarantee shall begin 2 years from the date of shipment from the manufacturer's plant. The life of the cells shall be considered at an end when the cells cannot be made to deliver 80 percent of their rated capacity when handled and maintained in accordance with the methods and procedures specified by the contractor's service manuals. Defective cells may be jumpered out of the battery

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as long as the remaining cells as a battery are capable of delivering 80 percent of the rated capacity; including any correction for temperature, before either the total battery voltage falls below 25.9 volts for a 15 cell battery (includes any jumpered cells) or 51.4 volts for a 30 cell battery (includes any jumpered cells) or before any cell falls below 1.74 volts.

6.2.3 The use or availability of spare (see 6.3.3) cells for replacement of failed cells in the battery set shall not alter the terms of this guarantee. Any cell shall be considered failed in the following cases:

(a) The cell fails to meet the guaranteed life provisions stated above.
(b) The cell limits the battery voltage below the minimum final voltage specified in table I and has to be removed in order to raise the overall battery final voltage.
(c) The cell is internally shorted.
(d) The cell develops leaks, thus causing grounds in the system.

6.2.4 In case of failure to meet the guaranteed provisions contained in 6.2.1 through 6.2.3, it is understood and agreed that the contractor will indemnify the Government on a prorata basis (proportionately) for months of service or equivalent cycles not obtained. (If P is the purchase price of the cells, n the number of months not obtained, Q the number of equivalent cycles not obtained, m the months of guaranteed service life, and E the number of guaranteed equivalent cycles, the smaller of the two numbers P X n/m or P X Q/E shall be used.)

6.2.5 At the option of the Government, the amount to be paid the Government may be assessed as a cash indemnity, or may be applied to reduce the cost of any other similar DSV main storage battery cells which the Government may contract to purchase from the contractor. It is understood and agreed that a cash indemnity will only be assessed when, in the opinion of the Government, experience indicates that the cell or type of cell manufactured under this contract by the contractor is unsuitable for use in DSV main storage batteries.

6.2.6 The guarantee shall also cover latent defects found in cells inspected by the shipyard during preparation of a battery for installation. The defective cells shall be rejected and replaced at no cost to the Government or the Government shall be reimbursed, unless the defective cells are repaired by the contractor to the Government's satisfaction. If repaired by the shipyard to expedite preparation of the battery, the manufacturer agrees to pay the Government the costs incurred by the shipyard.

6.3 Definitions. The definitions specified in 6.3.1 through 6.3.4 are applicable to this specification.

6.3.1 Cell. A cell is an element installed in a jar with a cover in place. The cell may be completely dry, with dry plates and separators, or may be filled with electrolyte.

6.3.2 Set. A set is the total number of cells grouped as indicated in 3.1.1 necessary to assemble one complete battery installation in a DSV.

6.3.3 Spares. Spare cells are cells supplied in excess of the cells constituting a set. There shall always be 9 spares for each lot of 90 cells (see 3.1.1 and 6.4).

6.3.4 Accessories. The accessories shall be furnished for each lot of 90 cells to be assembled on a ship as a battery (see 3.5).

6.4 Total hardware to be delivered. The total number of cells, materials, and parts to be delivered per ship set shall be the sum of the following:

(a) Battery cells - 90
(b) Spare cells - 9.
(c) Quality conformance cells - 6.
(d) Replacement electrolyte entrainment eliminator - 9.
(e) Accessories (see 3.5).

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
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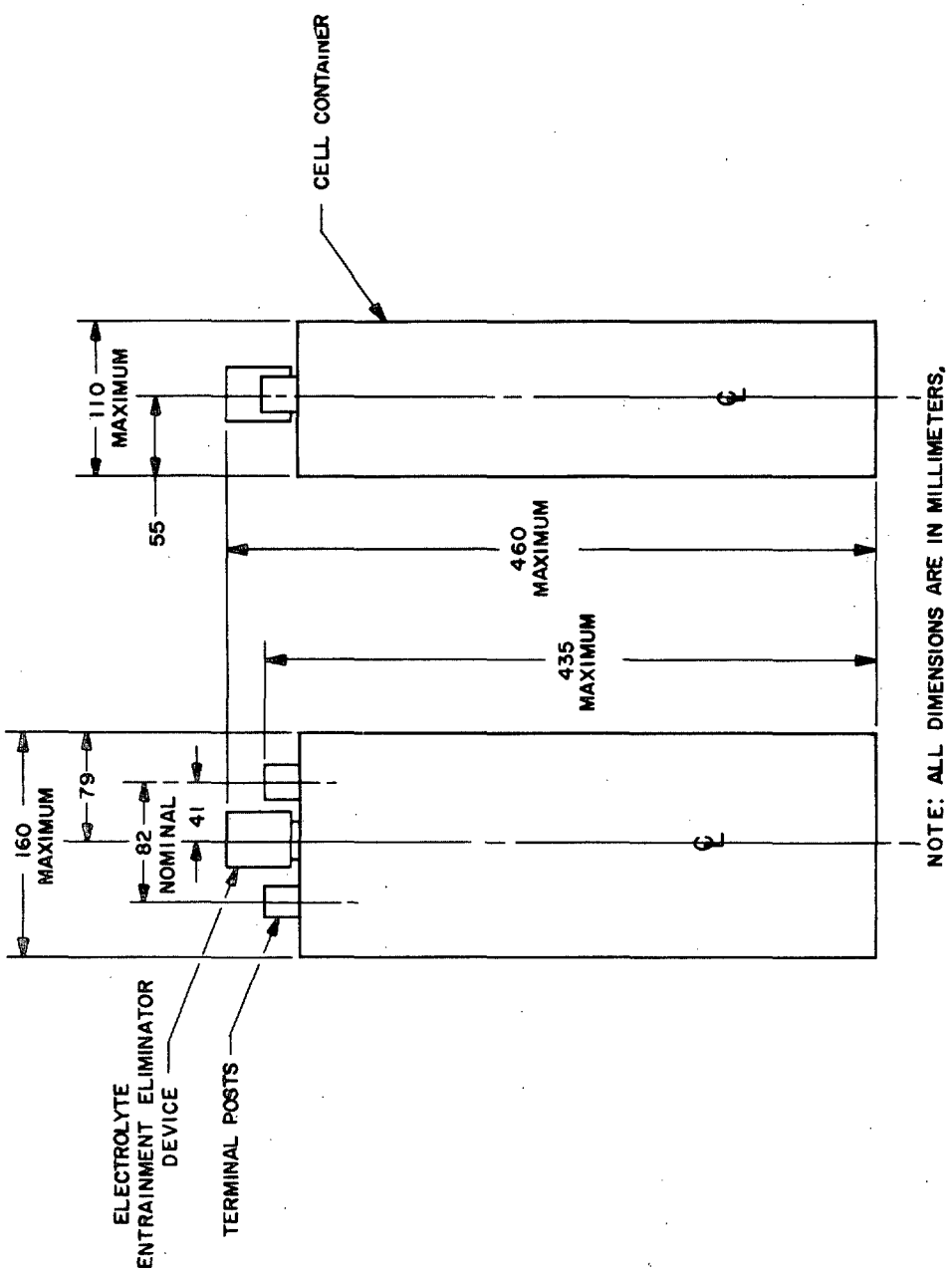


FIGURE 1. Basic dimensions of cell.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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